

Categorization of Nutritional Deficiencies in Plants With Random Forest

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Abstract- Agriculture supports to greater extent for the means of living in India, especially in rural areas. But generally the crop production attained by farmers would be much below the optimal production. The main reason for the crop production gap is due to the lack of essential soil nutrients and irrigation in the agricultural farms. To escalate the crop production, it is essential to balance the chemical elements or nutrients present in the soil with varying parameters of soil like the pH and soil moisture. Crop productivity can be increased to optimum level by efficient soil nutrient management. In case of Nutrient deficiencies, visual symptoms will appear on the leaf. This paper put forwards a method to identify the nutrient deficiencies of cabbage leaves by making use of visual symptoms appearing on the leaves by Classification. Seven types of deficiencies N, P, K, Ca, B, Zn and Mg are considered in this study. The proposed study consists of creation and pre- processing of a set of images consisting of nutrient deficient and healthy leaves of cabbage, feature extraction and by using Random Forest performing multi class classification of nutrient deficient leaves. The paper focuses on recognizing the visual indications of nutritional deficiency and thereafter classification.

Keywords-Balance, Classification, Nutritional Deficiency, Random Forest, Visual Symptoms

I. INTRODUCTION

Agriculture is the main source of living for nearly 60% of the population in India and it plays a major role in the GDP. Agriculture depends on adequate amount of sunlight, water and the absorption of nutrients by plants. The precise detection of nutritional deficiencies are essential for maintaining the ideal plant growth and thereby enhancing the crop production. Visual inspection is the initial indicator to decide whether the crop has nutritional deficiency or not.

In crops every nutrient will indicate a deficiency with different separate symptoms. Nutrient reduction depends on the rainfall, fertilizer usage, terrain and the land management practices.

Observable indications of nutrient deficiencies and toxicities would be the first indication to show that a particular area or location within a field has a nutritional problem. Identifying and noticing these visual symptoms is very important for taking the corrective action. By efficiently managing the nutrient contents present in the soil deficiencies occurring in plants can be controlled. Additionally, more evaluations can be done with exhaustive leaf and soil sampling.

The functions of essential nutrients in plants and its deficiency symptoms are given below.

Nitrogen is mainly essential for the building of proteins in plants. It helps in improving the crop yield as well as it is essential for the proper growth of plants. When there is Nitrogen deficiency, colour of leaf turns from green to yellow. The leaf and the veins will become uniformly yellow in colour. In case of acute deficiency, the colour of leaf becomes whitish yellow. Mostly nitrogen is absorbed in the soil as ammonium nitrate[1]

Phosphorous is needed for cell generation and division and in due course results in the growth of plants. Necrosis of older leaves and the undersized plants with short petiole are the notable symptoms. Plants take phosphorous as phosphate from the soil. Phosphorous absorption depends on the pH of the soil.[1].

Potassium has crucial role in the plant growth. It is necessary for activating many enzymes during metabolic reactions, controlling the opening and closing of stomata for regulating water loss and for the transportation of sugar .It plays a vital role in photosynthesis as well as carrying the photo assimilates all over the plant[1].Tip burn of leaves, necrosis between the veins of the leaves and the drooping of leaves and branches are the main symptoms. Potassium is absorbed in the form of ions from the soil.

Calcium deficiency results in brown spots of tip burn in young leaves, curling of leaves and diminutive growth of plants.

Boron deficiency causes a reduction in the growth of young leaves, flowers, buds, tip of roots points. Overall growth of the plant will be diminished [20].

Zinc deficiency results in interveinal yellowing of leaves called chlorosis. In cabbage it is exhibited as pale yellow green colour of leaf.

Magnesium is mobile in plants and the symptoms appear initially in the lower leaves. Chlorosis starts from the margin of the leaf and progresses to the middle and later the colour changes to brownish red . Big veins will be green in colour.

Earlier farmers were facing great trouble to identifying the nutritional deficiency causing to the crops which lead to great economic loss. If a farmer can detect these deficiencies early and apply appropriate action then it can save the crops from getting damaged and prevent the economic loss. Machine Learning and image processing techniques can be applied to come up with quick and precise results[2].

For considering an element as an essential element three points are to be noted: 1) No other element can replace its function 2) The element will be required throughout its life cycle 3) The element will be directly involved with the growth of plant.

This paper proposes an approach to identify the nutritional deficiencies, utilizing image processing and machine learning approaches. Deficiencies listed above are considered in this paper. The important aspects of this work are 1) Creating a data set of nutrient deficient leaves and healthy leaves. 2) Examining whether classification based on Random Forest can identify the nutrient deficiency 3) Estimating the performance of this approach

II. LITERATURE SURVEY

This section discusses the survey of different methods used for identifying the visual symptoms occurring in plants.

Susanto et al [3] developed computational intelligence vision sensing system for the estimation of nutrients in wheat leaves. Color features of the leaves images taken from the field under different conditions of light are examined. Deep sparse extreme learning machines combined with genetic algorithm is used to normalize plant images. Mean, variance, skewness, and kurtosis are taken as moments of color distribution and nutrients are estimated with these.

Zhiqiang et al [4] identified three steps to improve the Soil Available Nutrients estimation accuracy. Initially the rapid-nutrient assimilation method was used to optimize the crop growth simulation process. Compared with the ensemble Kalman filter. Rapid Nutrient Assimilation method showed an better performance for estimating nitrogen, phosphorus, and potassium

Ukrit et al [5] used convolutional neural network for identifying plant deficiencies. The network is trained to identify whether the plant shows any symptom for the deficiency. Image is divided into blocks and given to a set of CNNs. Every CNN is trained for detecting the particular symptoms for that particular deficiency. The responses are combined to one by making use of multilayer perceptron to get the response. Experiments were done on Vigna mungo (black gram).

Padmashri et al [6] Authors devised a system by considering maize leaf images deficient in nitrogen, phosphorous and potassium. The data set is prepared with nutrient deficient leaf images. The method was non-invasive way of detecting nutrient deficiency in plants and used Inception V3 CNN.

Sui R et al [7] Identified the nitrogen status in the soil for the cotton crop, by a sensing frame work developed in the ground. It consists of optical sensor, an ultrasonic sensor, and a data-acquisition and processing unit. Nitrogen status was predicted by making use of artificial neural network. Artificial neural network uses the sensor data. Accuracy was ninety percentage.

Imran et al [8] used aeroponics in their research. In this plants are grown in growth chamber. Nutrient solution is used instead of soil and nutrient mist is injected at regular intervals. Various parameters like electrical conductivity, carbon dioxide concentration pH of soil etc. are measured in regular intervals. Wireless sensors are used for the detection.

Achmad et al [9] used fuzzy logic to give recommendations for sugar cane essential nutrients in the soil. Fuzzification is done on soil moisture, pH, growth phase, and nutrient element.

Arvin et al [10] measured Nitrogen, Potassium and Phosphorus elements present in the soil, moisture and temperature for the growth of paddy by using sensors. Sensor data is analyzed and on the basis of data farmers could choose the suitable rice variant for cultivation

Goutham K et al [11] conducted their research on paddy crop. The authors measured the nutrients present in soil in real time and send the data to farmers. This helps the farmers to identify the type of paddy crop to be grown for a particular period. Experiments were conducted by measuring nitrogen, phosphorous and potassium in the soil. NPK values are classified as good, low and poor based on its percentage values in the soil.

In this paper the general sequence of image processing based nutrient deficiency extraction is explained initially and then in the latter part classification of the nutrient deficiencies are explained.

The relevant literature survey's indicated that Random Forest could be used for identifying nutritional deficiencies based on the symptoms appearing on leaves.

III. IMAGE PROCESSING & CLASSIFICATION

A. Acquire Image

Images of plant leaves can be captured either from the field with complex and complicated background [12] or with white background under controlled conditions [13]. Images may be captured through different devices like mobile phones, CCD cameras etc. Quality of image captured is based on the camera and the orientation of the principal axis of lens to the plane of the leaf. Accuracy of nutrient deficiency detection depends on the image capturing process because it is the input for training [18]

Two thousand images of the leaves and plant with a resolution of 3546 X 4608 pixel were taken using VIVO V21E. Three hundred and fifty images of healthy plants and remaining images with nutrient deficiency were acquired.

B. Pre-process the Image.

The image captured may contain noise either due to improper brightness, undesirable objects, poor contrast etc. So the image captured will go through noise removal

Acquired image is resized and noise is removed from the image. In median filter, abrupt change in pixel value is replaced with median of the neighboring values of the pixel. On the contrary, bilateral filter removes noise by replacing the abrupt change in intensity value of the pixel by weighted average of the intensity values of neighboring pixel values. In image processing applications Gaussian filter will blur images and removes noise based on Gaussian distribution. Laplacian filters are derivative filters used to detect abrupt change in edges of the image.

C. Feature Extraction

Feature extraction can be considered as a dimensionality reduction. Features can be grouped into shape, texture, colour etc. Length, width, perimeter, area etc. can be considered as features of shape while contrast, homogeneity,

correlation, etc. can be considered as features of Texture type.

Nutrient deficient leaves will have brown spots, patches, brown color at the tip, In order to identify these, features should be extracted from the image .For that color space of the image may be divided into bins. Each bin correspond to particular color [18]. Pixels are placed in the bin which is very close to its color.

D. Edge Detection

Uniformly green colored leaves are healthy leaves. Edge detection is done to identify the patches and spots. Sobel filter[19] is used for edge detection. In a gray scale image, the sudden change in intensity of the image is treated as edge. Sobel operator makes use of kernel and convolution operation to identify the edges. Two kernels for approximating the change in intensity values of pixels in x-direction and y-direction are used. Approximating the change in intensity values of pixels means approximating the gradient of intensity values.(G_x -horizontally in x-direction and G_y - vertically in y-direction).

$$\text{Magnitude of Gradient for pixel}(x,y)=\text{Sqrt}[G_x^2 + G_y^2]$$

$$\text{For pixel}(x,y) \text{ direction of gradient } \theta = \text{atan}(G_y/G_x)$$

E. Classification

Classification is a supervised learning approach in which the class of new observations or data is identified based on the training data. In this paper Random Forest, an ensemble learning used for classifying the nutrient deficiencies. Random Forest can deal with large datasets and it gives higher accuracy as compared with decision tree.

IV. DATASET USED

Cabbage grown in balanced nutrient solution and nutrient deficient solution was studied. Nitrogen, Potassium, Phosphorous, Calcium, Boron, Zinc and Magnesium deficient solutions were used for the experiment [23].In order to maintain the concentration of nutrient deficient solution, every week the solutions were changed. After one month the images of the leaves and plant were taken every day using VIVO V21E. Two thousand leaf and plant images were taken with a resolution of 3546 X 4608 pixel.

Manually segmented leaves of cabbage are shown in the Fig.1

Identifying the nutrient deficiency analysis involves

1. Leaf or plant analysis of the crop
2. Find the correlation of crop yield with any crop related characteristics like the amount of nutrients present in the plant.
3. Calibrating the correlation result in terms of crop yield [8]
4. Interpret the result.

A. Training, Test and Validation Data Set

In order to develop a model using machine learning three distinct data sets are required.

1) *Training Data-This is the initial data used by the network to learn the hidden parameters automatically.*

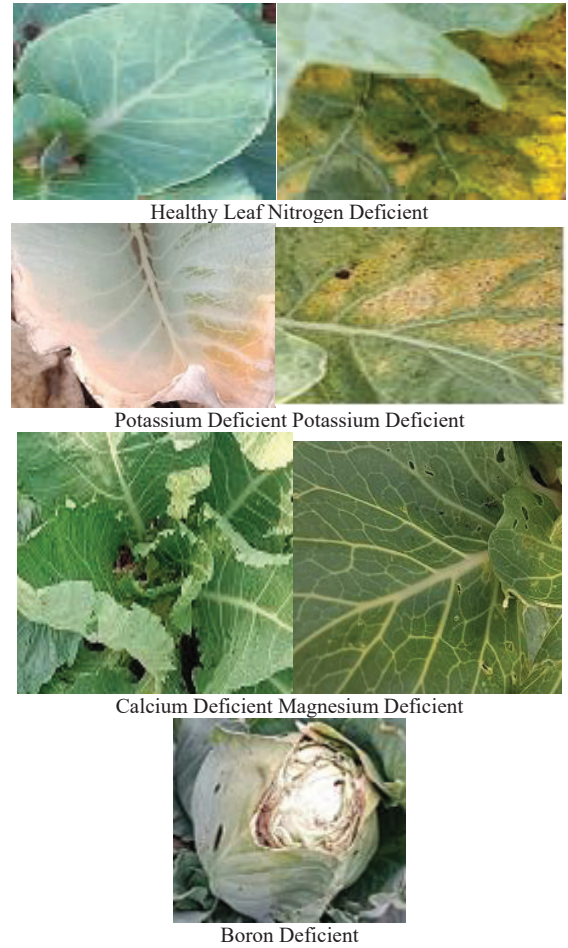


Fig. 1. Nutrient Deficient Symptoms

2) *Test Data-After the model is built using training data set , unseen data is required to test the model, called the test data.*

3) *Validation Data set-There are some hyper parameters which cannot be learned automatically during training. These parameters like learning rate, size of batch needs to be adjusted manually using the validation set.*

Images are converted to uniform fixed size. Machine learning requires enormous amount of data in order to get good accuracy for the model. Data is augmented by performing various operations to get large set of images from the existing data. Augmentation creates more images from existing images and it helps in increasing the accuracy of model.

In order to identify nutrient deficiency, data annotation of images [15] raises the query to identify the best part of image to send to the network ie. the entire leaf or part of leaf or the whole plant.

V. SYSTEM ARCHITECTURE

This section outline the research design and the method for identifying and classifying nutritional deficiencies. For this study Nitrogen, Phosphorous, Potassium, Calcium, Boron, Zinc and Magnesium deficiencies were considered.

This study made use of images of leaves with deficiencies in Nitrogen, Phosphorous, Potassium, and Calcium in Cabbage[8][22]. Two thousand leaf images of Cabbage were considered. The nutritional deficiencies

classified by the model were manually verified by the experts [5].

Random Forest is an ensemble learning used for classification and regression. Decision trees are created on data models and the nutrient deficiencies are estimated from each of the decision tree. Aggregation is then generated using CART(Classification and ReIntegration Tree Algorithm)[24]. Random forest gives high accuracy as the majority vote form each decision tree is used for classification.

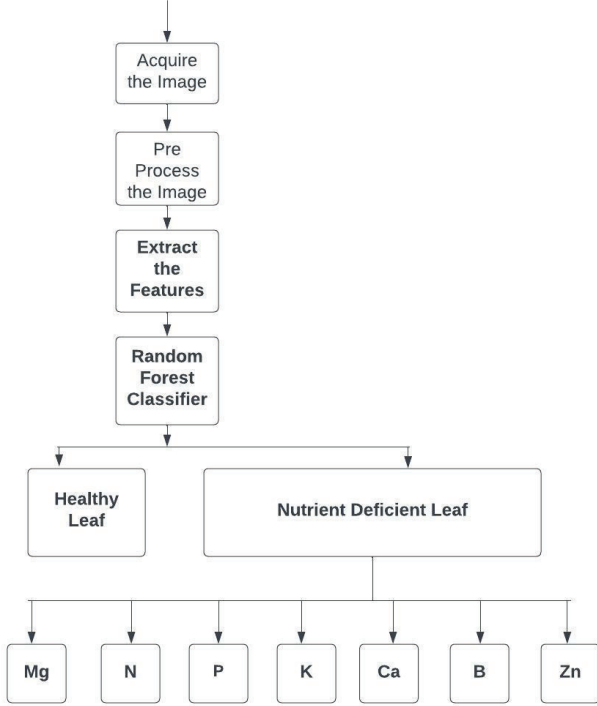


Fig. 2. Proposed System Architecture

VI. METHODOLOGY

- 1 Image Collection.- Acquire the images of healthy and nutrient deficient leaves.
- 2 Preprocess the image- Convert the collected images to uniform size. Augment the image to get large set of images by doing different operations on image like rotation, width shift, flip, fill etc.
- 3 Extract features and annotate the image Labels like colour, spots, tip brown, curly leaf ,brown patches ,yellow colored veins are considered.
- 4 Training and Test data Data set was divided into training(80%) and test data(20%).
- 5 Model building using Random Forest classifier. Classify the leaves into healthy and nutrient deficient leaves.

VII. PERFORMANCE EVALUATION

The study uses the metrics Precision and Recall for the decision support.[21].Performance of the system was evaluated using Precision, Recall [5][16][17] and F-measure[5][14].Twenty agriculture experts were also asked to identify the nutritional deficiencies manually from the

image set. Prediction of the agriculture experts is then compared with that of the proposed model [5].

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (1)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (2)$$

$$\text{F - measure} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Average of the prediction results of agriculture experts were taken for the calculation of Precision, Recall and F-measure.

Using the same metrics the performance of the proposed model was also evaluated. The proposed method gave more accurate deficiency categorization as compared with prediction results of agricultural experts.

TABLE I. EVALUATION SUMMARY

Class	Accuracy	
	Random Forest	Agriculture Expert
Healthy leaf	96.8	48
Nitrogen Deficient	95.7	28
Phosphorous Deficient	94.3	26
Potassium Deficient	94.8	23
Calcium Deficient	95.8	27
Boron Deficient	96.1	22
Magnesium Deficient	95.2	24
Zinc Deficient	96.3	22

VIII. DISCUSSION

The nutritional deficiency identification task by the agricultural expert was difficult task. Initial difficulty was that some deficiency symptoms overlap for different nutrients. For example necrosis can occur for Magnesium as well as for Potassium deficiency. Second difficulty was due to the vast variation of the symptoms caused by the deficiency of the particular nutrient ie. variations within a particular class.

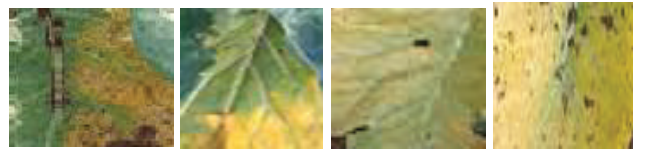


Fig. 3. Variations within the same class

The appearance of symptoms will be visible initially and will become noticeable as the days progress.

IX. CONCLUSION AND FUTURE WORK

The study shows the feasibility of using Random Forest classifier for identifying and categorizing the nutrient deficiencies present in the cabbage leaves. Random Forest classifier based identification is definitely preferable over identification by human experts. Overlapping symptoms for

the same nutrient, progressive appearance of symptoms and variations within the class impose some challenges in the nutrient deficiency identification.

The next step is to use the sensor technology for nutrient deficiency identification of the soil[23]. From the sensor signal, it is possible to identify the amount of nutrients present in the soil.

Sensor to sensor networking requires routing and synchronization. Physical layer design need to be upgraded for attaining energy saving. Sensing technology helps in enhancing the crop production and results in optimum crop management. In the agricultural field, this will help in attaining sustainability.

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